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Position-aware attentional neural network for review-based recommendation

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ABSTRACT

In e-commerce, user reviews have become one of the most effective sources to improve recommendation accuracy, as they provide rich semantic signals about user preferences and item characteristics. However, most existing review-based systems recommend items mainly by exploiting lexical and semantic cues, while overlooking the positional distribution of tokens and sentences. In practice, reviews exhibit clear spatial regularities, and incorporating such patterns can enhance the ability to capture key information. To address this limitation, we propose **Position-aware Attentional Neural Network (PAAN)**, which explicitly treats position as a central signal in review. Specifically, PAAN introduces a position-aware excitation module that integrates explicit learnable positional embeddings with contextual features to recalibrate attention and highlight informative tokens. By modeling where important information appears in reviews, PAAN improves the extraction of discriminative preference signals, especially under sparse user-item interactions where review content provides crucial auxiliary evidence. Moreover, a dual cross-attention mechanism further aligns interactions between user and item representations through enhanced review features, ensuring that relevance is learned in a structurally coherent and personalized manner. Extensive experiments demonstrate that our approach consistently outperforms existing review-based recommendation methods in rating prediction tasks. Compared to baseline methods, our model achieves an average improvement of 2.49% in MSE and 4.86% in MAE across five highly sparse datasets: Digital Music, Health and Personal Care, Home and Kitchen, Movies and TV, and Yelp. The code is available at https://github.com/YuanpengJiang/PAAN_Recommender.

1. Introduction

With the continuous expansion of information volume, people are facing a serious problem of information overload in the vast amount of content. To alleviate this problem, recommendation systems have become indispensable by efficiently filtering and ranking candidate items through multidimensional modeling of user preferences and item characteristics. Early collaborative filtering (CF) methods (Hu et al., 2008) and matrix factorization (MF) methods (He et al., 2017; Koren, 2008) mainly infer latent user preferences from rating matrices or interaction records. With the development of neural recommendation, graph-based models such as NGCF (Wang et al., 2019) and LightGCN (He et al., 2020) further represent users and items as graph nodes, enabling high-order collaborative signals to be propagated through neighborhood aggregation. More recently, self-supervised graph recommendation models, such as SGL (Wu et al., 2021) and SimGCL (Yu et al., 2022), have been proposed to learn more robust user and item representations from interaction graphs. Nevertheless, these methods mainly depend on user-item

interactions and graph structures, while overlooking the rich semantic information contained in user-generated reviews.

In fact, reviews often reflect personal opinions and detailed product descriptions, making them a valuable complementary signal in recommendation (Chen et al., 2018). Unlike numerical ratings that provide only coarse-grained signals, reviews contain fine-grained linguistic evidence about sentiment, intent, and aspect-level satisfaction. Recent studies have therefore attempted to combine ratings and reviews for better user and item representation learning. For example, RI-GCN (Cai et al., 2022) models review-aware user-item interactions, while the pre-trained Transformer with hierarchical attention network (Yen et al., 2024) captures contextual review semantics. More recent works, such as APH (Liu et al., 2025) and ARRR (Zhang et al., 2025), further explore aspect-level review information through hypergraph modeling or rating-review representation fusion. Overall, these methods enhance the representation of user-item interactions by capturing semantic, aspect-level, or relational information from reviews.

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Fig. 1. Cross-domain review examples illustrating positional sentiment signals. The examples cover Home and Kitchen, Movies and TV, and Yelp reviews. Sentiment-rich phrases are highlighted with color: red for positive, blue for negative.

However, most existing models focus on global semantic modeling while neglecting the **structural position information** inherently present in review sequences. As shown in Fig. 1, the four review examples cover real products, movie, and local-service scenarios, suggesting that positional signals are not limited to a single review domain. In general, reviews may exhibit a discourse structure where the opening sentence often introduces the main opinion, the middle portion elaborates on specific features, and the closing sentence tends to summarize or convey emotional conclusions. Such a structural discourse pattern substantially influences the overall semantic direction of a review and thus holds significant representational value.

Although these examples share a broadly similar positional structure, the information carried by each position can differ across domains. Product reviews usually place functional judgments around item attributes and usage effects, movie reviews often embed opinions in discussions of plot, acting, or continuity, and Yelp reviews tend to organize preferences around service-process details such as waiting time, food quality, and final choice. Therefore, the value of positional modeling does not lie in assuming a fixed writing template, but in capturing how domain-specific evidence is distributed within a review. For instance, in Lauren’s one-star review, the opening phrase “What a cheap mess” and the closing remark “I would not recommend wasting your time though” jointly determine the negative stance, whereas mid-sentence phrases such as “Luckily” or “made good on the broken item” may weaken this signal under position-agnostic aggregation. Hence, explicitly introducing a position-aware module holds promise for uncovering the latent discourse structure within reviews and improving a model’s ability to distinguish and decouple sentiment signals.

Although prior works have applied CNNs, LSTMs, and attention mechanisms—which to some extent consider sequential order—these models typically encode position only implicitly. In deeper layers of the network, positional signals tend to be diluted, limiting their impact on semantic representation. Specifically, CNNs and LSTMs can capture local sequential dependencies at shallow layers, but their capacity to preserve positional structure diminishes at higher levels of abstraction. Recent Transformer-based recommenders, such as CL4SRec (Xie et al., 2022) and DuoRec (Qiu et al., 2022), further strengthen sequential dependency modeling, but they mainly target user behavior sequences rather than the discourse structure within review texts. In short, existing approaches only partially utilize positional signals, leaving structural information underexploited.

To address these limitations, we propose Review-based Position-aware Attentional Neural Network (PAAN), a novel recommendation model that explicitly integrates positional information into review modeling. PAAN consists of two key components. The position-aware (PA) excitation module combines explicit learnable positional embeddings with contextual features. It recalibrates attention across different review segments and preserves structural cues that are often diluted in deep layers. The dual cross-attention (A) mechanism aligns review representations with user and item identities. In summary, these designs enable PAAN to capture both positional regularities and fine-grained semantic interactions. Accordingly, the contributions of this paper are as follows:

- We propose PAAN, a review-based recommendation framework that focuses on exploiting the positional representational capacity of reviews. To the best of our knowledge, PAAN is the first review-based recommendation model to systematically exploit structural positional signals at token-level.
- We introduce a position-aware excitation module for recommender systems. By combining explicit learnable positional embeddings with multi-statistic pooling and a feed-forward network, it adaptively generates position-sensitive attention weights, enhancing the model’s awareness of the structural layout in review content.
- We further design a dual cross-attention mechanism that aligns user and item reviews from both perspectives, effectively strengthening the interaction between review semantics and user/item ID embeddings to enrich representation quality.
- Extensive experiments on five real-world benchmark datasets demonstrate that PAAN significantly outperforms state-of-the-art models across various evaluation metrics, validating its effectiveness and robustness.

The rest of this paper is organized as follows. Section 2 provides a comprehensive overview of related work, highlighting recent advances in recommendation. Section 3 introduces the proposed methodology in detail, including the architecture design and key components of our model. Section 4 describes the experimental settings, evaluation metrics, and presents a thorough analysis of the empirical results. In the end, Section 5 summarizes our work.

2. Related work

In this section, we briefly review two research lines related to our work, namely semantic modeling of reviews for recommendation and representation learning with networks for recommendation.

2.1. Semantic modeling of reviews for recommendation

Text reviews serve as a crucial auxiliary signal in recommendation systems, exhibiting substantial value in semantic modeling (Wang et al., 2023). However, the effective utilization of review information depends on how these semantic signals are extracted and integrated. To date, research has predominantly focused on two main directions: **sentiment extraction** and **aspect extraction** (Bittencourt et al., 2025).

1) Sentiment extraction: These methods aim to mine users' emotional inclinations from review texts, thereby enhancing personalization and user satisfaction in recommender systems (Elahi et al., 2023). Representative works include Ashok et al. (2016) and Lei et al. (2016), which classify reviews into positive, neutral, and negative polarities using machine learning techniques and combined these with ratings to optimize ranking. Dong et al. (2020) propose emotion extraction using word-level and sentence-level attention mechanisms and incorporate the Gumbel Softmax activation function to compute probabilities at the emotion level. Zhang et al. (2021) combine sentiment labels to guide review representation learning and exploits sentiment information in user-item interactions to improve recommendation performance. Ao and Zhang (2023) incorporate sentiment features from reviews into collaborative filtering, combining matrix factorization and MLP to enhance recommendation accuracy. Despite these advances, such methods generally rely heavily on high-quality sentiment-labeled data, which is costly and often inconsistent, and they tend to overlook the internal structure of reviews and the importance of content located at different positions.

2) Aspect extraction: This line of research is based on the assumption that, within a specific review context (e.g., product reviews), user comments tend to center around meta-themes or aspects of the items. Users express their preferences through the aspects emphasized in their own reviews, while items are represented by aggregating aspects from reviews. Anand and Naorem (2016) propose methods to automatically detect aspects and corresponding opinions without labeled data, thereby enabling more precise extraction of feature-level user sentiments. However, aggregating all reviews without differentiation may obscure individual user preferences. To address this, Chin et al. (2018) employ attention mechanisms to extract aspects from individual user reviews, and Cheng's ALFM (Cheng et al., 2018) learns latent aspects from rating matrix factorization and combines them with aspects derived from user reviews, ensuring personalized preference modeling. Models like MA-GNNs (Zhang et al., 2023) integrate aspect-based sentiments from reviews into multiple user-item graphs and fuse them through a routing mechanism to capture fine-grained preferences. Furthermore, APH (Liu et al., 2025) learns item aspect performance through hypergraph modeling, while ARRR (Zhang et al., 2025) combines review and rating representations to capture user behavior patterns and aspect preferences. However, these signals are outside the scope of this paper, where we focus on uncovering the structural semantic organization embedded within review texts and improving both sentiment recognition and recommendation decision accuracy.

2.2. Representation learning with networks for recommendation

Neural networks have been widely adopted in recommender systems for their ability to learn complex and nonlinear representations. In review-aware recommendation, a key challenge is how to effectively capture semantic, structural, and sentiment features from textual reviews (Bittencourt et al., 2025). Existing neural architectures mainly follow two lines: **Graph Convolutional Networks (GCNs)** and **Convolutional Neural Networks (CNNs)**.

1) GCNs: These propagate information across user-item graph structures to learn effective representations via neighborhood aggregation. In rating-based tasks, GCNs have been widely adopted in models such as PinSage (Ying et al., 2018), NGCF (Wang et al., 2019), and LightGCN (He et al., 2020), which aggregate neighboring nodes through graph convolution operations to learn effective user and item representations. Recent

graph-based recommendation models further refine this line of research. DGA-IPR (Ye et al., 2023) reconstructs influence and preference relations for rating prediction, DMGNN (Song et al., 2022) learns user and item representations from multiple graphs through attention fusion, and E-GNN (Sheng et al., 2023) improves session-based recommendation by modeling global item transitions and local session patterns. Some GCN-based methods enrich review representations with auxiliary signals. RI-GCN (Cai et al., 2022) integrates ratings and reviews into a user-item bipartite graph to capture contextual interactions, while RGNN (Liu et al., 2023) constructs user/item-specific review graphs with type-aware attention and hierarchical pooling.

Beyond review-only modeling, recent work leverages external knowledge to ground and explain review signals. Liu and Miyazaki (2023) extract knowledge entities from reviews and use dual attention to capture knowledge-level user-item interactions, while Bastola and Shakya (2024) fuse reviews with knowledge bases to generate path-level explanations. Recent knowledge-enhanced models further extend this direction: IGAT (Elahi et al., 2025) incorporates knowledge graph information for item-specific recommendation, KGCAN (Elahi et al., 2024) models contextualized entity relations for user-specific recommendation, and Saadat et al. (2024) combine knowledge graph convolution with sentiment analysis for drug recommendation. IKSAR (He et al., 2025) further aligns interaction and knowledge graphs with user preference-aware item fusion. These methods enrich recommendation through external entities, relations, or reasoning paths. Although they partially overlap with our work in using review semantics, their main purpose is to expand the source of semantic evidence through knowledge graphs, whereas positional modeling focuses on the internal organization of reviews, namely where sentiment and preference cues appear within review sequences.

2) CNNs: These models rely on convolution operations to capture stable semantic representations within local contexts, making them well-suited for extracting salient phrases and word combinations from review texts. ConvMF (Kim et al., 2016) first introduced CNNs into recommender systems to extract semantic cues from reviews. DeepCoNN (Zheng et al., 2017) further encodes user and item reviews with dual channels. Later studies incorporate interaction data, pair-specific CNN features, factorization machines (Wu et al., 2019), and multi-domain encoding with domain discrimination (Choi et al., 2022). Although these CNN-based models demonstrate effectiveness, they tend to neglect structural variations in review content and the importance of different positions in influencing recommendation outcomes. General NLP encoders such as BERT (Devlin et al., 2019) preserve token order through positional embeddings, but they do not explicitly model the structural roles of different review positions in recommendation.

3. Methodology

In this section, we begin with the formal definition of the review-based recommendation task, followed by an overview of the PAAN architecture. We then elaborate on the embedding fusion strategy that integrates review text and positional signals, and present the design of the position-aware excitation module. Finally, we describe the dual cross-attention mechanism, which also incorporates the fusion strategy for modeling user-item interactions.

3.1. Problem formulation

Let $\mathcal{U} = \{u_1, \dots, u_{|\mathcal{U}|}\}$ and $\mathcal{I} = \{i_1, \dots, i_{|\mathcal{I}|}\}$ denote the user and item sets, respectively. We observe a subset $\Omega \subseteq \mathcal{U} \times \mathcal{I}$ of interactions. For each $(u, i) \in \Omega$, a rating $r_{u,i} \in \mathbb{R}$ is given, along with a user-side review $\mathcal{R}_u = \{r_u^1, \dots, r_u^L\}$ and an item-side review $\mathcal{R}_i = \{r_i^1, \dots, r_i^L\}$ of fixed length L . The task is to learn a function $f: \mathcal{U} \times \mathcal{I} \rightarrow \mathbb{R}$ such that $\hat{r}_{u,i} = f(u, i)$ approximates $r_{u,i}$ for both seen and unseen pairs.

We represent each review as contextual token embeddings $\mathbf{E}^u$ and $\mathbf{E}^i$ with shape $\mathbb{R}^{L \times D}$. To inject order information, we denote learnable

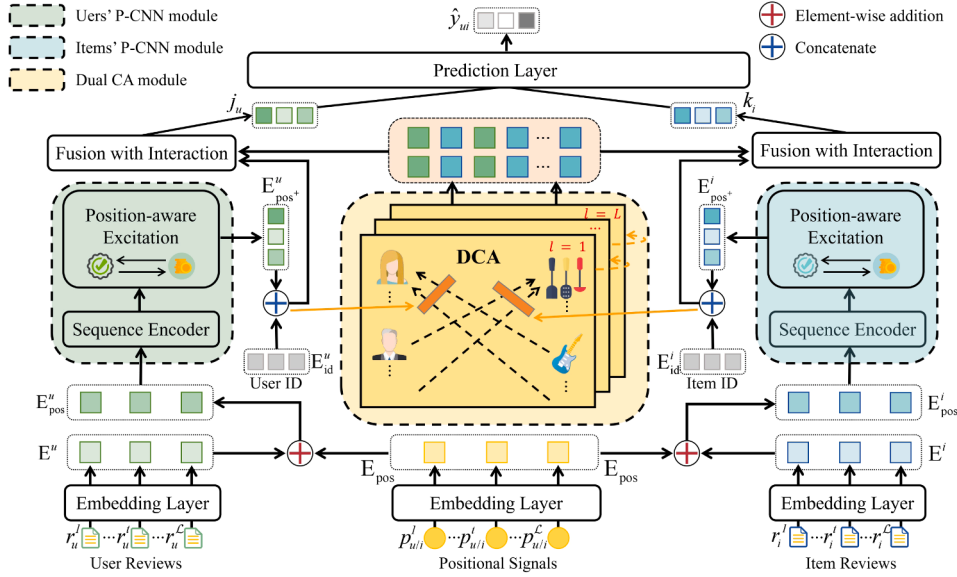


Fig. 2. An overview of the architecture of the proposed PAAN model, which integrates position-aware excitation and dual cross-attention mechanisms to jointly model review semantics, positional information, and user–item interactions for rating prediction.

positional embeddings by $\mathbf{E}_{pos} \in \mathbb{R}^{L \times D}$ and refer to the resulting position-enriched inputs as $\mathbf{E}_{pos}^u$ and $\mathbf{E}_{pos}^i$. A sequence encoder is used to obtain intermediate representations, denoted by $\mathbf{H}^u$ and $\mathbf{H}^i$, which capture local semantic patterns. User and item IDs are represented by $\mathbf{E}_{id}^u, \mathbf{E}_{id}^i \in \mathbb{R}^{d_{id}}$ and will later be fused with text-derived features.

A position-aware excitation module $\text{PAE}(\cdot)$ is applied to $\mathbf{E}_{pos}^u$ and $\mathbf{E}_{pos}^i$ to recalibrate token-level features. Specifically, channel descriptors p_{avg} and p_{max} are obtained via pooling, concatenated, and passed through a gated transformation, producing elementwise weights $\mathbf{A} \in \mathbb{R}^{L \times D}$ for excitation. To capture fine-grained semantic alignment, we employ the dual cross-attention between $\mathbf{G}^u$ and $\mathbf{G}^i$. Let $\text{Attn}(Q, K, V)$ be multi-head scaled dot-product attention. Before dual cross-attention, the position-aware review features are first concatenated with user/item ID embeddings to form $\mathbf{G}^u$ and $\mathbf{G}^i$. In this process, each token representation in $\mathbf{G}^u$ attends to all tokens in $\mathbf{G}^i$ to produce an aligned sequence $= \mathbf{Z}_{i \rightarrow u}$, and vice versa for $\mathbf{Z}_{u \rightarrow i}$. After temporal pooling, we obtain interaction-aware vectors $\mathbf{j}_u, \mathbf{k}_i \in \mathbb{R}^d$, which are fed into a predictor $h(\cdot)$ to output $\hat{y}_{u,i}$.

3.2. Framework overview

As illustrated in Fig. 2, PAAN processes user-side and item-side review sequences together with their positional signals through a series of clearly defined stages. The model progressively transforms these inputs into an estimated rating $\hat{y}_{u,i}$. The layered architecture is summarized as follows:

- **Input & Embedding Fusion.** It explicitly injects positional signals $\mathbf{E}_{pos}$ into textual representations $\mathbf{E}^u$ and $\mathbf{E}^i$, which are often ignored in conventional review encoders but are essential for modeling discourse structure. To better capture positional information in reviews, we treat $\mathbf{E}_{pos}$ as a learnable embedding rather than a fixed one, allowing it to be continuously updated to highlight key positional signals. Meanwhile, the resulting position-enriched inputs $\mathbf{E}_{pos}^u$ and $\mathbf{E}_{pos}^i$ preserve the original embedding dimension D .
- **Sequence Encoder & Position-aware Excitation.** It leverages the strength of convolutional operations and positional modeling to efficiently capture local semantic signals while emphasizing informative positions. It applies a one-dimensional convolutional encoder to $\mathbf{E}_{pos}^u$ and $\mathbf{E}_{pos}^i$ to transform local token features, complementing global attention at the token level. Two-dimensional convolutions are deliberately avoided, since reviews are sequential rather than spatial data.

The resulting feature maps $\mathbf{H}^u$ and $\mathbf{H}^i$ are then recalibrated through a position-aware excitation mechanism. By integrating multi-pooling statistics with positional embeddings via adaptive gating, this module explicitly accounts for positional importance and produces enhanced features $\mathbf{E}_{pos+}^u$ and $\mathbf{E}_{pos+}^i$, allowing sentiment-rich segments to be emphasized rather than treated uniformly.

- **Dual Cross-Attention & Interaction.** It uses a dual cross-attention mechanism to capture asymmetric but complementary dependencies between user and item reviews. This design allows the model to reflect fine-grained user-item interactions while keeping representations balanced. The recalibrated features $\mathbf{E}_{pos+}^u$ and $\mathbf{E}_{pos+}^i$ are first concatenated with ID embeddings $\mathbf{E}_{id}^u$ and $\mathbf{E}_{id}^i$. They are then fed into the dual cross-attention module to produce interaction-aware vectors $\mathbf{j}_u$ and $\mathbf{k}_i$.
- **Prediction.** The interaction-aware vectors $\mathbf{j}_u$ and $\mathbf{k}_i$ are aggregated and fed into the predictor $h(\cdot)$ to estimate the final rating $\hat{y}_{u,i}$.

3.3. Embedding fusion of reviews and positions

Given a review sequence of length L , the textual tokens for users and items are first mapped into dense word embeddings. Let $\mathbf{E}^u \in \mathbb{R}^{B \times L \times D}$ and $\mathbf{E}^i \in \mathbb{R}^{B \times L \times D}$ denote the user-side and item-side word embeddings respectively, where B is the batch size and D the embedding dimension.

Simultaneously, learnable positional embeddings $\mathbf{E}_{pos} \in \mathbb{R}^{L \times D}$ are introduced to encode token positions in the sequence. These positional embeddings are broadcasted across the batch and added element-wise to the corresponding word embeddings:

$$\mathbf{E}_{pos}^u = \mathbf{E}^u + \mathbf{E}_{pos}, \quad \mathbf{E}_{pos}^i = \mathbf{E}^i + \mathbf{E}_{pos} \quad (1)$$

Here, $\mathbf{E}_{pos}^u$ and $\mathbf{E}_{pos}^i \in \mathbb{R}^{B \times L \times D}$ represent the position-enriched embeddings for user and item reviews respectively. This fusion integrates textual and positional signals at the embedding level, serving as the input for subsequent sequence encoding.

3.4. Position-aware excitation module

The Position-aware Excitation (PAE) module is inspired by Hu et al. (2018), which recalibrates channel-wise features using global statistics. Different from the original channel-excitation design, ordinary feature recalibration is not sufficient for review modeling, because the informativeness of review tokens is closely related to where they appear in the

Table 1

Position-aware excitation module.

Algorithm 1: Position-aware Excitation Module

Input: Encoded review representation $\mathbf{H}^* \in \mathbb{R}^{B \times L \times D}$, where $\star \in \{u, i\}$; reduction ratio r ; temperature coefficient τ ; residual coefficient α

Learnable parameters: PAE positional embedding $\mathbf{E}_{\text{pos}}^{\text{pae}} \in \mathbb{R}^{L \times D}$; pooling logits $\beta = \{\beta_{\text{avg}}, \beta_{\text{max}}\}$; gating parameters $\mathbf{W}_1, \mathbf{W}_2, \mathbf{b}_1, \mathbf{b}_2$

Output: Recalibrated position-aware representation $\mathbf{E}_{\text{pos}^+}^* \in \mathbb{R}^{B \times L \times D}$

Function PAE($\mathbf{H}^*$):

// Step 1: weighted token summary

$p_{\text{avg}} \leftarrow \text{Mean}_d(\mathbf{H}^*), \quad p_{\text{max}} \leftarrow \text{Max}_d(\mathbf{H}^*)$

$p \leftarrow \text{WeightedPool}(p_{\text{avg}}, p_{\text{max}}; \beta, \tau)$

// Step 2: position-conditioned coupling

$\tilde{\mathbf{E}}_{\text{pos}}^{\text{pae}} \leftarrow \text{Broadcast}(\mathbf{E}_{\text{pos}}^{\text{pae}}, B), \quad \tilde{\mathbf{p}} \leftarrow \text{Expand}(p, D)$

$\mathbf{F} \leftarrow \text{Concat}(\tilde{\mathbf{p}}, \tilde{\mathbf{E}}_{\text{pos}}^{\text{pae}}), \quad \mathbf{F} \in \mathbb{R}^{B \times L \times 2D}$

// Step 3: position-aware gating

$\mathbf{M} \leftarrow \text{ReLU}(\mathbf{F}\mathbf{W}_1 + \mathbf{b}_1)$

$\mathbf{A} \leftarrow \sigma(\mathbf{M}\mathbf{W}_2 + \mathbf{b}_2), \quad \mathbf{A} \in \mathbb{R}^{B \times L \times D}$

// Step 4: residual recalibration

$\mathbf{E}_{\text{pos}^+}^* \leftarrow \mathbf{H}^* \odot \mathbf{A} + \alpha \mathbf{H}^*$

return $\mathbf{E}_{\text{pos}^+}^*$

Function Main:

$\mathbf{E}_{\text{pos}^+}^u \leftarrow \text{PAE}(\mathbf{H}^u), \quad \mathbf{E}_{\text{pos}^+}^i \leftarrow \text{PAE}(\mathbf{H}^i)$

review discourse. As illustrated in Fig. 1, preference-bearing cues may occur in opening opinions, middle feature descriptions, or closing emotional conclusions, and position-agnostic aggregation may weaken or mix these signals. Therefore, PAE is tailored to sequential review representations by jointly considering token-level statistical cues and explicit positional signals, enabling the model to recalibrate review features according to position-dependent informativeness rather than relying only on global semantic strength.

Before entering PAE, the position-enriched embeddings obtained in Eq. (1), namely $\mathbf{E}_{\text{pos}}^u$ and $\mathbf{E}_{\text{pos}}^i$, are first passed through the sequence encoder to capture local semantic patterns:

$$\mathbf{H}^u = \text{Encoder}(\mathbf{E}_{\text{pos}}^u), \quad \mathbf{H}^i = \text{Encoder}(\mathbf{E}_{\text{pos}}^i), \quad (2)$$

where $\mathbf{H}^u, \mathbf{H}^i \in \mathbb{R}^{B \times L \times D}$ denote the encoded user/item side review feature maps; the PAE procedure with $\mathbf{H}^*$ as input is summarized in Algorithm 1.

1) Weighted Multi-Pooling Fusion: Given the encoded representation $\mathbf{H}^* = [\mathbf{h}_{b,l,d}^*] \in \mathbb{R}^{B \times L \times D}$, where $\star \in \{u, i\}$, PAE first extracts two complementary token-level statistics along the channel dimension. Specifically, average pooling captures the overall channel response at each position, while max pooling highlights the most salient activation:

$$p_{\text{avg}}^{(b,l)} = \frac{1}{D} \sum_{d=1}^D h_{b,l,d}^*, \quad p_{\text{max}}^{(b,l)} = \max_{1 \leq d \leq D} h_{b,l,d}^*, \quad (3)$$

where $b \in \{1, \dots, B\}$ and $l \in \{1, \dots, L\}$ denote the batch index and token position, respectively. To avoid treating the two statistics with fixed importance, we introduce learnable pooling logits $\beta = \{\beta_{\text{avg}}, \beta_{\text{max}}\}$ and compute a temperature-scaled weighted fusion:

$$p^{(b,l)} = \sum_{m \in \{\text{avg}, \text{max}\}} \frac{\exp(\beta_m / \tau)}{\sum_{m' \in \{\text{avg}, \text{max}\}} \exp(\beta_{m'} / \tau)} p_m^{(b,l)}, \quad (4)$$

Here, $\tau > 0$ serves as a temperature parameter controlling the sharpness of the weighting distribution. Smaller values of τ produce highly peaked distributions that emphasize the most dominant components, while larger values yield smoother and more evenly distributed weights.

In this way, the module adaptively combines average-pooled and max-pooled token summaries, providing a more flexible statistical descriptor for subsequent position-aware excitation.

2) Position Embedding Broadcasting and Feature Concatenation: Position-specific information is embedded by a learnable matrix:

$$\mathbf{E}_{\text{pos}}^{\text{pae}} = [\mathbf{e}_1^{\text{pae}}]_{l=1}^L \in \mathbb{R}^{L \times D} \quad (5)$$

To align the positional embeddings with the batch structure, each position vector $\mathbf{e}_l^{\text{pae}}$ is broadcast across all batch indices $b \in \{1, \dots, B\}$, yielding $\tilde{\mathbf{E}}_{\text{pos}}^{\text{pae}} \in \mathbb{R}^{B \times L \times D}$, where each element along the batch dimension shares the same positional value.

At the same time, the pooled scalar $p^{(b,l)}$ obtained from the weighted multi-pooling step is expanded along the feature dimension so that each channel $d \in \{1, \dots, D\}$ receives the same weight, producing $\tilde{p}^{(b,l,d)} = p^{(b,l)}$.

Finally, the expanded pooled vector and the broadcasted positional embedding are concatenated along the channel dimension to form the position-augmented feature:

$$\mathbf{F}^{(b,l)} = [\tilde{p}^{(b,l,1)}, \dots, \tilde{p}^{(b,l,D)}, e_1^{\text{pae},1}, \dots, e_l^{\text{pae},D}]^T \in \mathbb{R}^{2D} \quad (6)$$

This concatenation enriches the pooled statistical features with explicit positional signals, enabling the downstream gating network to leverage positional context during feature recalibration.

3) Adaptive Gating and Residual Recalibration: While positional embeddings can enhance feature representations, some position signals may be irrelevant or noisy, reducing semantic clarity. To address this, we introduce an adaptive gating mechanism that selectively modulates position-augmented features before integration into the final representation.

Specifically, the concatenated features $\mathbf{F}^{(b,l)}$ are first transformed by a two-layer feedforward network, enabling non-linear interactions between position embeddings and excitation weights. The first transformation applies a dimensionality reduction to enhance computational efficiency and encourage compact latent interactions:

$$\mathbf{M}^{(b,l)} = \text{ReLU}(\mathbf{F}^{(b,l)}\mathbf{W}_1 + \mathbf{b}_1), \quad (7)$$

$$\mathbf{W}_1 \in \mathbb{R}^{2D \times \frac{D}{r}}, \quad \mathbf{b}_1 \in \mathbb{R}^{\frac{D}{r}},$$

where r is the reduction ratio controlling the compression of intermediate channels. The second transformation restores the dimensionality and applies a sigmoid activation $\sigma(\cdot)$ to produce gating weights $s^{(b,l,d)}$ in the range $(0, 1)$, effectively serving as a soft mask over each token-channel response:

$$s^{(b,l,d)} = \sigma(\mathbf{M}^{(b,l)}\mathbf{W}_2 + \mathbf{b}_2), \quad (8)$$

$$\mathbf{W}_2 \in \mathbb{R}^{\frac{D}{r} \times D}, \quad \mathbf{b}_2 \in \mathbb{R}^D.$$

These gates allow the model to adaptively determine the contribution of position-aware features per token and per channel. Gates close to zero indicate that positional information is suppressed, while values near one signal a high reliance on position semantics.

Finally, the recalibrated feature map is obtained by element-wise modulation of the encoded representation $\mathbf{H}^*$ with the learned gates, followed by a residual connection scaled by a learnable parameter α :

$$\mathbf{E}_{\text{pos}^+}^* = \mathbf{H}^* \odot \mathbf{A} + \alpha \mathbf{H}^*. \quad (9)$$

This residual design preserves original features, preventing semantic degradation when the gating mechanism misjudges position signals, and stabilizes gradient flow via a direct feature path. To avoid excessive suppression or amplification in early training, α is initialized to a small value (e.g., 0.1 to 1.0), yielding a near-identity mapping. Gating weights use a bias b_2 such that $\sigma(b_2) \approx 0.5$, ensuring balanced initial contributions from original and position-augmented features. This controlled setup enhances convergence stability and prevents overfitting to spurious position signals.

3.5. Dual cross-attention mechanism

This design employs dual-perspective attention from the user and item sides, unlike DAML's global-local scheme (Liu et al., 2019). It enables semantic alignment and mutual refinement of feature embeddings, effectively modeling asymmetric interactions in review texts. Let $\mathbf{U}, \mathbf{I} \in \mathbb{R}^{B \times L \times D}$ denote the user-side and item-side representations fed into the dual cross-attention module, respectively, where B , L , and D are the batch size, sequence length, and hidden dimension. The cross-attention operation from user to item is formulated as follows:

$$\mathbf{Z}_{u \rightarrow i} = \text{Concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}_O \quad (10)$$

where each attention head is computed by

$$\text{head}_h = \text{Softmax}\left(\frac{Q_u^{(h)} (K_i^{(h)})^\top}{\sqrt{d_h}}\right) V_i^{(h)} \quad (11)$$

with

$$Q_u^{(h)} = \mathbf{U} \mathbf{W}_Q^{(h)}, \quad K_i^{(h)} = \mathbf{I} \mathbf{W}_K^{(h)}, \quad V_i^{(h)} = \mathbf{I} \mathbf{W}_V^{(h)} \quad (12)$$

Here, $Q_u^{(h)}, W_K^{(h)}, W_V^{(h)} \in \mathbb{R}^{D \times d_h}$ are the learnable projection matrices for the h th head, H denotes the total number of heads, and $d_h = D/H$ represents the dimensionality of each head. $\mathbf{W}_O \in \mathbb{R}^{D \times D}$ is the output projection matrix consolidating multi-head outputs.

Symmetrically, the attention from item to user is defined by swapping the roles of $\mathbf{U}$ and $\mathbf{I}$:

$$\mathbf{Z}_{i \rightarrow u} = \text{CrossAttn}(\mathbf{I}, \mathbf{U}, \mathbf{U}) \quad (13)$$

Since the above operations perform token-level alignment in both directions, we further analyze the computational cost of the dual cross-attention module. For a batch size B , sequence length L , hidden dimension D , number of heads H , and $d_h = D/H$, one cross-attention direction requires $O(BLD^2)$ time for linear projections and output transformation, and $O(BHL^2d_h) = O(BL^2D)$ time for attention score computation and value aggregation. Thus, the time complexity of one direction is $O(BLD^2 + BL^2D)$. Since the proposed dual cross-attention computes both user-to-item and item-to-user alignments, the total cost is $O(2(BLD^2 + BL^2D))$, which preserves the same asymptotic order. The space complexity is $O(BHL^2 + BLD)$, mainly from storing multi-head attention matrices and intermediate sequence representations.

To derive compact representations suitable for subsequent prediction layers, we perform average pooling over the sequence length dimension, aggregating the cross-attended features:

$$\mathbf{j}_u = \frac{1}{L} \sum_{i=1}^L \mathbf{Z}_{u \rightarrow i}^{(:, :, :)} \quad (14)$$

$$\mathbf{k}_i = \frac{1}{L} \sum_{i=1}^L \mathbf{Z}_{i \rightarrow u}^{(:, :, :)} \quad (15)$$

Dropout is applied to the attention weights to regularize training and mitigate overfitting. After obtaining the interaction-aware vectors $\mathbf{j}_u$ and $\mathbf{k}_i$, the model combines them by concatenating their element-wise product with a learned interaction:

$$\mathbf{z} = [\mathbf{j}_u \odot \mathbf{k}_i; \varphi(\mathbf{j}_u, \mathbf{k}_i)] \quad (16)$$

where $\varphi(\cdot)$ is a learnable function such as a linear layer. This fusion captures both explicit matching signals and higher-order user-item relations for prediction.

4. Experiments

In this section, we conduct extensive experiments to comprehensively evaluate the effectiveness of the proposed PAAN model. The experiments aim to answer the following research questions:

Table 2

Statistics of five public datasets.

Dataset	#Users	#Items	#Reviews	Density
Music	5541	3568	64,706	0.327%
Health	38,609	18,534	346,355	0.048%
Kitchen	66,519	28,237	551,682	0.029%
Movies	123,960	50,052	1,697,533	0.027%
Yelp	99,011	56,441	2,762,098	0.049%

- **RQ1:** How does PAAN perform compared to state-of-the-art review-based recommendation models?
- **RQ2:** How effective is the proposed position-aware modeling in improving robustness and prediction accuracy compared with standard Transformer-based sequence modeling?
- **RQ3:** How do the position-aware excitation and dual cross-attention mechanisms contribute to the model's performance?
- **RQ4:** How do different hyperparameters, such as the number of attention heads, kernel size and reduction ratio, affect the model's performance?
- **RQ5:** How well can PAAN capture and visualize the influence of relative review positions on sentiment intensity, and what insights can be drawn from case studies?

Datasets. We evaluate our model on five publicly available Amazon datasets, including four Amazon datasets¹ and the Yelp Dataset.² The Amazon datasets include Digital Music (shortened as **Music**), Home and Kitchen (**Kitchen**), Health and Personal Care (**Health**), and Movies and TV (**Movies**). Each dataset is used in its 5-core version, ensuring that both users and items have a minimum of five reviews, which helps reduce extreme sparsity. For Yelp, we follow a stricter 10-core filtering strategy, retaining only users and items with at least ten reviews. These datasets cover diverse domains and exhibit varying degrees of rating distribution, user-item interaction density, and positional sentiment variation. Such diversity creates a comprehensive benchmark for evaluating our model's robustness in different application contexts. To provide an overview of data characteristics, we report in Table 2 the number of users, items, reviews and sparsity.

Evaluation Metrics. Following Kokubo and Sugiyama (2022), we adopt two widely used evaluation metrics, namely Mean Squared Error (MSE) and Mean Absolute Error (MAE), to comprehensively assess the prediction accuracy of our model:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_{ui} - y_{ui})^2 \quad (17)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_{ui} - y_{ui}| \quad (18)$$

where N is the total number of predictions, $\hat{y}_{ui}$ denotes the predicted rating, and y_{ui} is the corresponding ground-truth rating.

Baseline. To verify the effectiveness of the PAAN method, we compare it against the following twelve baselines:

- **NeuMF** (He et al., 2017) combines generalized matrix factorization with a multi-layer perceptron to model non-linear user-item interactions, providing a flexible framework for collaborative filtering.
- **DeepCoNN** (Zheng et al., 2017) encodes user and item reviews through two parallel convolutional networks, then combines the learned features for predicting user ratings.
- **NARRE** (Chen et al., 2018) introduces a review-level attention mechanism to identify and emphasize more informative reviews, improving both prediction accuracy and interpretability.

¹ <http://jmcauley.ucsd.edu/data/amazon>.

² <https://business.yelp.com/data/resources/open-dataset/>.

- *ALFM* (Cheng et al., 2018) incorporates aspect information derived from reviews into latent factor modeling, allowing recommendation to reflect fine-grained user preferences.
- *DAML* (Liu et al., 2019) employs a dual-attention mutual learning strategy to capture complementary local and global semantic features from textual reviews.
- *CARL* (Wu et al., 2019) integrates CNN-based review representations with attention-based fusion, enabling more expressive modeling of user-item interactions.
- *LightGCN* (He et al., 2020) is a simplified graph convolutional approach for collaborative filtering that focuses solely on neighborhood aggregation without feature transformation.
- *RGCL* (Shuai et al., 2022) leverages review texts and the user-item interaction graph in a contrastive learning framework to enhance recommendation robustness.
- *KAAN* (Liu & Miyazaki, 2023) exploits knowledge entities extracted from reviews to model user-item relations at the knowledge level, enabling clearer interpretation of recommendations.
- *KEGC* (Bastola & Shaky, 2024) draws on external semantic knowledge to enrich user, item, and interaction representations within a graph neural network, supporting both accurate prediction and explanation generation.
- *ARRR* (Zhang et al., 2025) fuses review-based and rating-based representations for aspect-level recommendation, where ratings are used to capture user behavior patterns and item quality.
- *APH* (Liu et al., 2025) constructs an aspect-aware hypergraph from reviews and models the relationships among users, items, aspects, and sentiment polarities for review-based recommendation.

These baselines can be categorized into four groups according to their primary modeling techniques: (1) the methods which utilize textual reviews to enhance rating prediction, i.e., DeepCoNN, NARRE, DAML, CARL and RGCL. (2) the methods which focus on modeling user-item interactions without review semantics, i.e., NeuMF and LightGCN. (3) the methods which model fine-grained aspect-level information from reviews, i.e., ALFM, ARRR and APH. (4) the methods which incorporate external or review-derived knowledge graphs to improve interpretability and predictive accuracy, i.e., KAAAN and KEGC.

Parameter Settings. We perform standard preprocessing for all datasets. Review texts are first tokenized and normalized, after which the vocabulary is limited to the 5,000 most frequent tokens. Each review document is padded or truncated to a fixed length L , and a pre-trained BERT model (Devlin et al., 2019) is used to extract token-level contextual representations from the last hidden layer, resulting in embeddings

of size $\mathbb{R}^{L \times 768}$ for each review document. For each dataset, we randomly split the data into training, validation, and test sets with proportions of 80%, 10%, and 10%, respectively. In our experiments, we set the batch size to 64 and use the Adaptive Moment Estimation (Adam) (Kingma & Ba, 2014) optimizer with an initial learning rate of 1×10^{-3} and a weight decay of 1×10^{-4} . The learning rate is decayed by a factor of 0.8 every 5 epochs. Dropout is fixed at 0.3 in the sequence encoder, cross-attention, and feature interaction modules. The fully connected layer in the transformer module has a dimension of 32, and ID embeddings are set to 128. The residual weight α in the position-aware excitation module is tuned within [0.1, 1.0]. All experiments are conducted on a workstation equipped with an NVIDIA GeForce RTX 4090 GPU, ensuring efficient training and reproducibility.

4.1. Overall performance (RQ1)

Table 3 shows the performance of the proposed PAAN model compared to the baselines across five datasets. Each experiment is repeated five times with different random seeds, and paired t -tests are conducted between PAAN and the best-performing baseline under each metric. Except for MSE on Music, all improvements of PAAN pass the significance test at $p < 0.05$. We have the following main observations from the results:

First, models that exploit review information generally outperform those that rely only on ratings, and this advantage becomes especially evident in sparse datasets. Rating-only methods such as LightGCN and NeuMF remain competitive on denser collections (e.g., Music), where LightGCN achieves the best performance among baseline models, reducing MSE by 1.25% and MAE by 2.06% compared with DeepCoNN. However, these gains are rather limited when contrasted with the pronounced declines observed on sparse domains: for instance, on Movies, LightGCN's MSE and MAE increase by 8.54% and 7.08%, respectively, relative to DeepCoNN. A similar tendency can also be observed on Yelp, where NeuMF and LightGCN are clearly inferior to review-enhanced baselines. These results indicate that the effectiveness of rating-only approaches is highly sensitive to dataset characteristics, whereas review-based models leverage supplementary semantic cues that enable more robust preference modeling. This highlights the critical role of user reviews in capturing nuanced user interests and item properties, and points to the necessity of continued research into review-enhanced recommendation paradigms.

Second, models that enhance rating predictions with review information show clear improvements over earlier methods that either ignore reviews or use them in a limited manner. For example, KEGC, ARRR, and

Table 3

Overall performance comparison on five datasets. The best baseline results are underlined, and the proposed method results are in **bold**. The performance improvement of PAAN (Improv.) is defined as $\text{Improv.} = \frac{\text{the best baseline result} - \text{Our result}}{\text{the best baseline result}}$. Statistical significance is examined by a paired t -test, where * denotes $p < 0.05$ compared with the best baseline.

Dataset	Music		Health		Kitchen		Movies		Yelp	
Metrics	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
DeepCoNN	0.7974	0.6690	1.0897	0.7758	1.1252	0.7812	1.0907	0.7920	1.2206	0.8770
NARRE	0.7926	0.6655	1.0841	0.7958	1.1317	0.8074	1.1086	0.8182	1.2239	0.8825
DAML	0.8055	0.6598	1.0757	0.8017	1.1191	0.7967	1.0892	0.8038	1.3366	0.9285
CARL	0.7917	0.6649	1.0906	0.7965	1.1391	0.8135	1.2539	0.8833	1.3596	0.9340
RGCL	0.7907	0.6617	1.0748	0.7841	1.1246	0.8008	1.0834	0.7896	1.3363	0.8849
NeuMF	0.7889	0.6691	1.1350	0.8285	1.1689	0.8310	1.2366	0.8728	1.4785	0.9628
LightGCN	<u>0.7874</u>	<u>0.6552</u>	1.0783	0.7877	1.1342	0.8094	1.1838	0.8481	1.3881	0.9428
ALFM	0.8649	0.7011	1.1751	0.8523	1.1972	0.8485	1.3368	0.9132	1.3856	0.9411
ARRR	0.8891	0.6964	1.1606	0.7795	1.1493	<u>0.7737</u>	1.0787	0.7847	1.2237	0.8835
APH	0.8102	0.6631	<u>1.0650</u>	0.7775	1.1392	0.8055	<u>1.0706</u>	<u>0.7824</u>	1.2501	<u>0.8708</u>
KAAN	0.8057	0.6650	1.0715	0.7886	1.1179	0.8000	1.1290	0.8143	1.3127	0.9185
KEGC	0.8040	0.6700	1.0666	<u>0.7637</u>	<u>1.1174</u>	0.7838	1.0769	0.7900	1.2714	0.8943
PAAN	0.7729	0.6281*	1.0452*	0.7233*	1.0975*	0.7280*	1.0356*	0.7318*	1.1756*	0.8491*
Improv.(%)	1.84	4.14	1.86	5.29	1.78	5.91	3.27	6.47	3.69	2.49

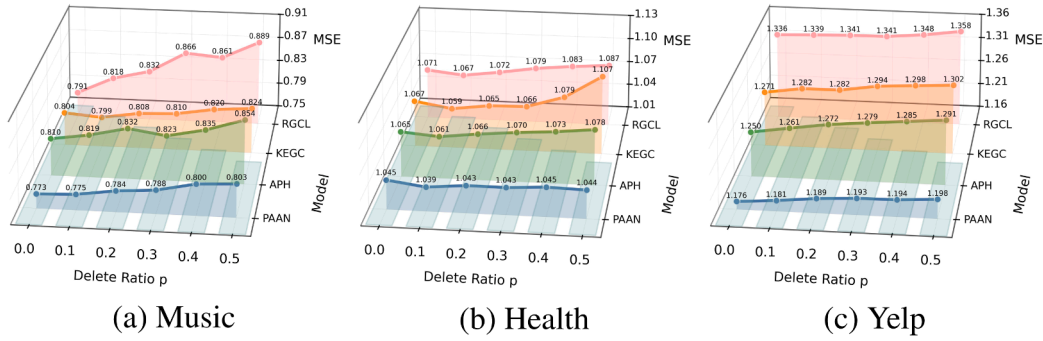


Fig. 3. Robustness evaluation of competing models across token masking levels on three datasets.

APH consistently outperform traditional review-based methods such as DeepCoNN and NARRE across most datasets. Notably, the aspect-aware baselines ARRR and APH also achieve competitive results on several datasets, indicating that fine-grained aspect modeling is beneficial for review-based recommendation. These results illustrate that incorporating review information—via aspect-level representation learning (ARRR and APH), or external semantic enrichment (KEGC)—enhances the model’s ability to capture nuanced user–item interactions. Overall, this suggests that effectively leveraging textual reviews can meaningfully improve recommendation accuracy.

Third, the proposed PAAN consistently achieves the best performance across all five datasets, surpassing the second-best baseline by **1.78%–3.69%** in MSE and **2.49%–6.47%** in MAE. These results demonstrate the effectiveness of PAAN, which models user–item interactions from complementary perspectives. The superiority of our approach mainly stems from two components: 1) the position-aware excitation module, which captures the impact of sentence positions in reviews and selectively emphasizes informative segments, and 2) the dual cross-attention mechanism, which models fine-grained interactions between user and item representations, allowing the model to better align review semantics with rating prediction. It is worth noting that the improvements are particularly pronounced on the **Health, Kitchen, and Movies** datasets. Moreover, PAAN also achieves consistent improvements on the cross-domain **Yelp** dataset, with 3.69% and 2.49% gains in MSE and MAE, respectively, over the best baseline. This observation suggests that the proposed method is also well-suited for alleviating the challenges posed by sparse data scenarios.

4.2. Position-aware modeling analysis (RQ2)

We examine position-aware modeling from two perspectives: robustness under partially degraded review information and effectiveness beyond standard Transformer-based sequence modeling. Specifically, we evaluate whether PAAN can preserve useful preference signals under token deletion and whether position-aware excitation offers additional gains over conventional positional encoding.

To evaluate the robustness of our model, we conduct token-level deletion tests on three datasets, i.e., Health, Music, and Yelp, with varying deletion ratios p . Specifically, we select three strong baseline models, namely APH, KEGC, and RGCL, which achieve competitive performance among the compared baselines, and report their results together with PAAN in Fig. 3. This setting allows us to examine the robustness of PAAN against relatively strong competitors rather than weak baselines. The deletion ratios are set to $[0.0, 0.1, 0.2, 0.3, 0.4, 0.5]$, where tokens are randomly removed to simulate incomplete or noisy review information in real-world scenarios.

As shown in Fig. 3, PAAN consistently achieves the lowest MSE across all deletion ratios on the three datasets, demonstrating its superior robustness under different levels of review degradation. On the Health dataset, PAAN remains highly stable, with MSE changing only

from 1.0452 at $p = 0.0$ to 1.0441 at $p = 0.5$, while APH, KEGC, and RGCL exhibit higher MSE values throughout the masking process. In particular, KEGC increases from 1.0666 to 1.1069, indicating that its performance is more sensitive to severe token loss. On the Music dataset, although all models suffer performance degradation as p increases, PAAN still maintains the best performance, increasing from 0.7729 to 0.8034, whereas RGCL shows a much larger degradation from 0.7907 to 0.8895. A similar trend can also be observed on Yelp, where PAAN increases moderately from 1.1756 to 1.1983, while the competing baselines remain consistently worse and show larger absolute MSE values. These results demonstrate that PAAN can better preserve useful preference signals under incomplete review inputs, indicating strong robustness against token-level information loss.

To further analyze whether standard sequence modeling is sufficient for review-based recommendation, we introduce two Transformer-based baselines under the same dataset and token-level review setting. Specifically, Trans. denotes a standard Transformer encoder, while Trans.-BERT replaces the input embeddings with token-level contextual representations extracted by BERT³.

As shown in Table 4, incorporating token-level BERT embeddings into the Transformer architecture does not lead to consistent improvements. Trans.-BERT slightly improves over Trans. on Music, but it performs worse on Health, Kitchen, Movies, and Yelp in both MSE and MAE. This phenomenon suggests that stronger contextual token representations alone are not sufficient for review-based rating prediction. One possible reason is that BERT mainly enhances local contextual semantics, whereas the recommendation task also requires identifying which positions or review segments are more informative for user preference and item characteristic modeling. Directly feeding contextualized tokens into a generic Transformer may introduce redundant semantic details or domain-specific noise, especially when reviews are long, sparse, or heterogeneous.

PAAN outperforms both Transformer-based baselines across all datasets, suggesting that its gains do not come merely from sequence modeling or contextual token embeddings. Instead, the position-aware excitation module recalibrates review representations with position-sensitive signals, helping the model focus on informative segments and reduce the influence of less useful positional patterns. This supports the value of explicit positional excitation over conventional Transformer-style positional encoding for review-based recommendation.

4.3. Ablation study (RQ3)

To examine how different components of PAAN contribute to its effectiveness, we designed ablation studies by constructing two groups of model variants:

³ <https://huggingface.co/google-bert/bert-base-uncased>.

Table 4

Performance comparison with Transformer-based sequence modeling baselines. The symbols $\uparrow$ and $\downarrow$ denote improvement and degradation, respectively.

Dataset	Music		Health		Kitchen		Movies		Yelp	
Metrics	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Trans.	0.8166	0.6628	1.0712	0.7513	1.1085	0.7874	1.0646	0.7662	1.2130	0.8548
Trans.-BERT	0.8120 $\uparrow$	0.6579 $\uparrow$	1.0771 $\downarrow$	0.7534 $\downarrow$	1.1660 $\downarrow$	0.8311 $\downarrow$	1.0840 $\downarrow$	0.7718 $\downarrow$	1.2894 $\downarrow$	0.8853 $\downarrow$
PAAN	0.7729	0.6281	1.0452	0.7233	1.0975	0.7280	1.0356	0.7318	1.1756	0.8491

Table 5

Performance comparison of PAAN and its variants.

Dataset	Method	MSE	Improv.1	MAE	Improv.2
Music	PAAN	0.7729	–	0.6281	–
	PA ^{gr} AN	0.7782	0.68%	0.6536	3.90%
	PA ^a AN	0.7809	1.02%	0.6502	3.40%
	PA ^{all} AN	0.7908	2.26%	0.6342	0.96%
	PAA ^u N	0.7948	2.76%	0.6589	4.67%
	PAA ⁱ N	0.7834	1.34%	0.6388	1.68%
	PAA ^{all} N	0.7965	2.96%	0.6543	4.00%
Health	PAAN	1.0452	–	0.7233	–
	PA ^{gr} AN	1.0543	0.86%	0.7281	0.66%
	PA ^a AN	1.0580	1.21%	0.7708	6.16%
	PA ^{all} AN	1.0711	2.42%	0.7361	1.74%
	PAA ^u N	1.0549	0.92%	0.7337	1.42%
	PAA ⁱ N	1.0611	1.50%	0.7411	2.40%
	PAA ^{all} N	1.0729	2.58%	0.7311	1.07%
Kitchen	PAAN	1.0975	–	0.7280	–
	PA ^{gr} AN	1.0949	–0.24%	0.7638	4.69%
	PA ^a AN	1.0991	0.15%	0.7760	6.19%
	PA ^{all} AN	1.1088	1.02%	0.7682	5.23%
	PAA ^u N	1.1149	1.56%	0.7331	0.70%
	PAA ⁱ N	1.1107	1.19%	0.7390	1.49%
	PAA ^{all} N	1.1313	2.99%	0.7463	2.45%
Movies	PAAN	1.0356	–	0.7318	–
	PA ^{gr} AN	1.0414	0.56%	0.7698	4.94%
	PA ^a AN	1.0509	1.46%	0.7774	5.87%
	PA ^{all} AN	1.0472	1.11%	0.7820	6.42%
	PAA ^u N	1.0496	1.33%	0.7380	0.84%
	PAA ⁱ N	1.0407	0.49%	0.7456	1.85%
	PAA ^{all} N	1.0512	1.48%	0.7567	3.29%
Yelp	PAAN	1.1756	–	0.8491	–
	PA ^{gr} AN	1.1907	1.27%	0.8546	0.64%
	PA ^a AN	1.1905	1.25%	0.8529	0.45%
	PA ^{all} AN	1.2229	3.87%	0.8623	1.53%
	PAA ^u N	1.2119	3.00%	0.8609	1.37%
	PAA ⁱ N	1.2051	2.45%	0.8633	1.64%
	PAA ^{all} N	1.2274	4.22%	0.8663	1.99%

- PA^xAN: This category of variants focuses on ablations within the position-aware (PA) module. Specifically, PA^{gr}AN denotes the removal of the adaptive gating mechanism and residual connection within the positional sequence excitation module, PA^aAN represents the exclusion of the entire positional sequence excitation module while retaining the positional embeddings, and PA^{all}AN corresponds to removing both the positional sequence excitation module and the positional embeddings entirely.
- PAA^xN: This group of variants examines ablations applied to the attentional (A) module. In detail, PAA^uN indicates the deletion of the user-to-item cross-attention mechanism, PAAⁱN denotes the removal of the item-to-user cross-attention, and PAA^{all}N signifies the complete elimination of the dual cross-attention mechanism.

We report the results of ablation studies in Table 5.

Effectiveness of position-aware (PA) module. Compared with PAAN, the variant PA^{all}AN shows an average increase of 2.14% in MSE and 3.18% in MAE across datasets. This demonstrates that the position-aware excitation module is crucial, as it allows the model to more effectively capture sequential dependencies and contextual cues in reviews. In contrast, the other two variants (PA^{gr}AN and PA^aAN) yield only marginal changes. This indicates that positional embeddings alone provide some benefit, but their effect remains limited without excitation. Moreover, removing the gating and residual mechanisms makes the model overly dependent on raw positional signals, which fails to deliver stable improvements. The same trend can also be observed on Yelp, where removing all position-aware components increases MSE and MAE by 3.87% and 1.53%, respectively, further confirming the generalization ability of the PA module across heterogeneous review domains.

Effectiveness of attentional (A) module. For the attentional module, we observe a similar pattern. Removing only the user-to-item (PAA^uN) or item-to-user (PAAⁱN) branch leads to a clear drop. This demonstrates that each direction captures complementary cues that cannot be replaced by the other. The removal of both directions (PAA^{all}N) results in the most severe degradation, with an average increase of 2.85% in MSE and 2.56% in MAE, confirming the essential role of dual cross-attention. Interestingly, on the Music dataset, we observe that PAA^uN and PAAⁱN exhibit a more pronounced discrepancy between their MSE and MAE values, which suggests that user-item interactions in music reviews are inherently more diverse and asymmetric. On Yelp, the complete removal of dual cross-attention also leads to the largest MSE degradation among all attentional variants, with a 4.22% increase over PAAN. Overall, these results demonstrate that the attentional module is critical for modeling fine-grained and asymmetric dependencies between users and items.

4.4. Hyperparameter sensitivity analysis (RQ4)

This section investigates how major hyper-parameters affect model performance, focusing on the number of attention heads, kernel size, and the reduction ratio. Due to the space limitation, we only present sensitivity results on the Music, Health, and Movies datasets, while noting that the observations on other datasets are similar.

Varying the number of attention heads. In our model, we employ multi-head cross-attention to capture diverse interaction patterns between user and item review representations. To investigate its impact, we vary the number of heads in [1, 2, 3, 4, 5] and report the results in Fig. 4. From the results, we have the following observations. First, using only a single head significantly degrades the performance, indicating that a single attention score is insufficient to capture the diverse semantic relations between user and item reviews. Second, the best performance is consistently achieved when the number of attention heads is 3, suggesting that a moderate number of heads can provide complementary perspectives while avoiding redundancy. In contrast, larger numbers of heads do not bring further improvement and may even lead to overfitting. In terms of efficiency, using more heads also introduces additional attention-map computation and memory overhead,

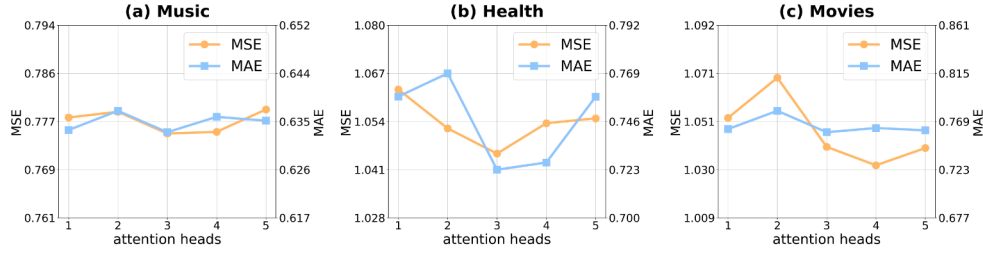


Fig. 4. Performance w.r.t number of attention heads.

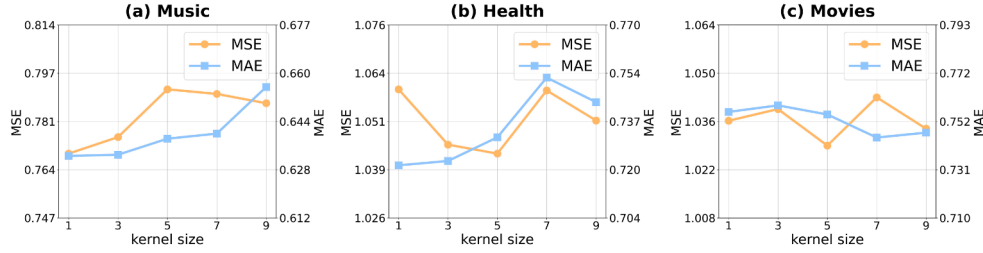


Fig. 5. Performance w.r.t kernel size.

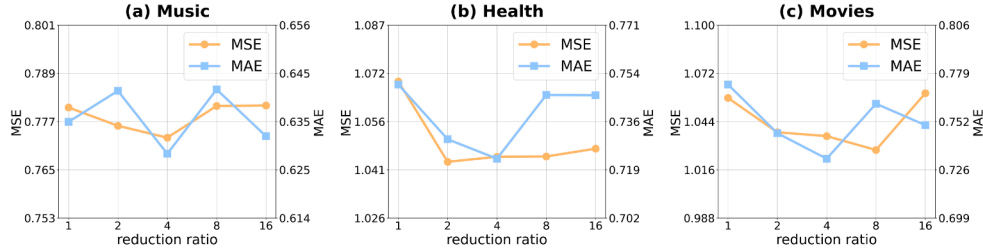


Fig. 6. Performance w.r.t reduction ratio.

although the total hidden dimension remains unchanged. Overall, we empirically find that setting the number of attention heads to 3 yields a good balance between expressiveness, generalization, and training efficiency.

Varying the size of kernel. To study its effect, we vary the kernel size in [1, 3, 5, 7, 9] and show the performance of our model in Fig. 5, from which we have some observations. On the Music dataset, the best performance is achieved with a kernel size of 1. For the medium-scale Health dataset, kernel sizes of 3 or 5 perform better, while on the large-scale Movies dataset, the optimal results appear with 5 or 7. This indicates that larger datasets require wider kernels to capture richer contextual information. In other words, as the amount of data and sequence diversity increase, a larger receptive field becomes essential for modeling long-range semantic patterns more effectively. However, increasing the kernel size also enlarges the convolutional cost and training time, and an overly wide kernel may introduce irrelevant neighboring information. Therefore, the kernel size should be selected according to the scale and semantic complexity of the dataset rather than being increased blindly.

Varying the ratio of reduction. We further study the effect of the reduction ratio in the position-aware module. The ratio controls how strongly the pooled features are compressed before generating excitation weights. Here we vary it in [1, 2, 4, 8, 16] and present the results in Fig. 6. The observations are clear. Very small ratios, such as 1 or 2, keep too many redundant signals and hurt the performance. Very large ratios, such as 8 or 16, compress the features too much and lead to weak representations. Across all datasets, the best results appear when the reduction ratio is set to 4. More specifically, $r = 4$ achieves the lowest or second-lowest MSE and MAE on the three reported datasets, showing that it provides a stable choice under different data scales and re-

view distributions. This shows that a moderate ratio strikes a good balance between removing noise and preserving useful information. From an efficiency perspective, larger ratios reduce the bottleneck dimension and computational cost, while overly large compression may discard position-sensitive signals.

4.5. Case study (RQ5)

In this section, we present a case study to examine how PAAN leverages positional signals and a dual cross-attention mechanism to capture sentiment cues and user-item interactions. First, we examine the Kitchen dataset as a whole. The sentiment scores shown in Fig. 7 are generated using the DistilBERT⁴ model, normalized on a 0–10 scale. Scores are computed by combining sentence-level and word-level contributions at a 4:1 ratio, incorporating neighbor weights.

As illustrated in Fig. 7, final sentiment intensity in the Kitchen dataset exhibits clear positional patterns: extreme sentiments tend to occur near the beginning and end of reviews, while mid-position sentences are often more neutral. This observation motivates a closer inspection of individual reviews to understand how PAAN leverages positional information.

Fig. 8 shows a detailed analysis of a review authored by Lauren within the Kitchen dataset. Pronounced negative sentiment is observed around relative positions 0.1 and 0.9, while distinct positive peaks emerge near positions 0.4 and 0.7. Models without the positional-awareness component (e.g., DAML) generally focus on sentences exhibiting strong emotional intensity but do not differentiate between positive and negative sentiments at specific positions. As a result, they tend

⁴ <https://huggingface.co/distilbert>.

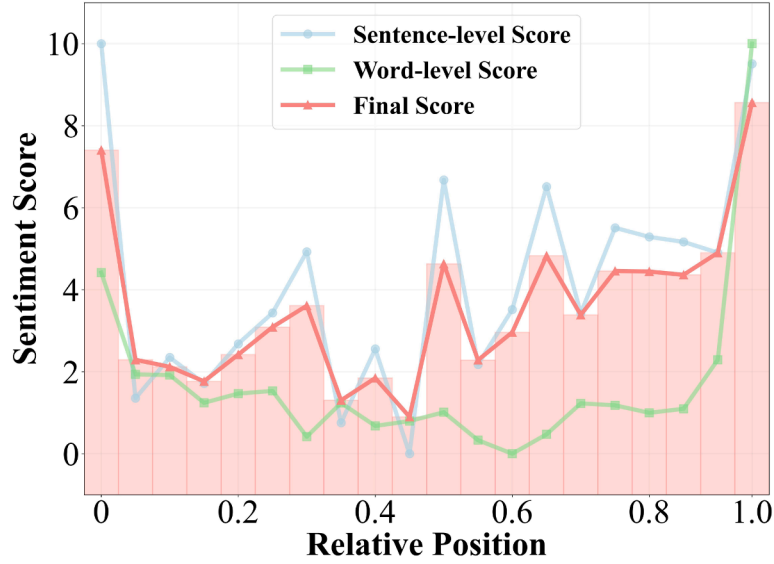


Fig. 7. Multi-level sentiment analysis on the Kitchen dataset, where higher scores correspond to stronger sentiment intensity regardless of polarity.

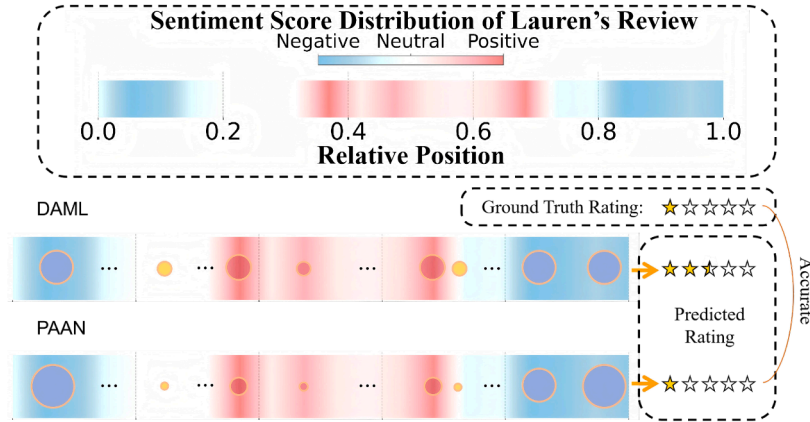


Fig. 8. Sentiment score visualization and rating predictions for Lauren's review (Kitchen dataset). Red indicates positive sentiment, blue indicates negative sentiment, and white represents neutral sentiment. Circle size reflects the attention weight assigned by PAAN to each sentence.

to treat all sentences uniformly, which can dilute the influence of key evaluative cues, particularly those located at the beginning or end of reviews. This indiscriminate attention may lead to misrepresentation of the overall sentiment and, consequently, reduce the accuracy of rating predictions. In contrast, PAAN adaptively downweights the contribution of mid-position segments, which are often less decisive, while amplifying the impact of extreme sentiments appearing at the beginning and end of reviews. This selective weighting emphasizes the negative expressions at the review boundaries that carry critical cues for overall judgment, allowing the model to more effectively capture the importance of different sentences and thereby improve the accuracy of rating predictions.

As shown in Fig. 9, PAAN uses a dual cross-attention mechanism to model interactions in both directions. The two attention maps reveal associations between user and item reviews, with stronger responses around sentiment words. The item to user map concentrates on a small group of negative cues, including “cheap”, “mess” and “broken”. By contrast, the user to item map is more dispersed and extends to utility related words such as “wasting” and “tool”.

4.6. Efficiency analysis

In addition to recommendation accuracy, computational efficiency is also an important factor for evaluating the practical applicability of

Table 6

Efficiency and recommendation effectiveness ranking of representative models. Time denotes the average epoch time in seconds; T-Rank denotes the efficiency rank based on Time; P-Rank denotes the average performance rank over ten dataset-metric combinations; E-Rank denotes the overall recommendation effectiveness rank.

Model	Time(s)	T-Rank	P-Rank	E-Rank
DeepCoNN	31.36	1	5.50	5
KEGC	93.89	2	5.10	2
RGCL	125.74	3	5.40	4
PAAN	190.85	4	1.00	1
APH	412.32	5	5.10	2

review-based recommendation models. Therefore, we further compare the average epoch time of representative models on the Health dataset and report the corresponding efficiency and effectiveness rankings in Table 6. We use Health as the representative dataset for time measurement because the same efficiency order, i.e., T-Rank, is consistently observed on the other datasets.

As shown in Table 6, DeepCoNN, as a classic model that jointly learns user and item representations from reviews, achieves the fastest train-

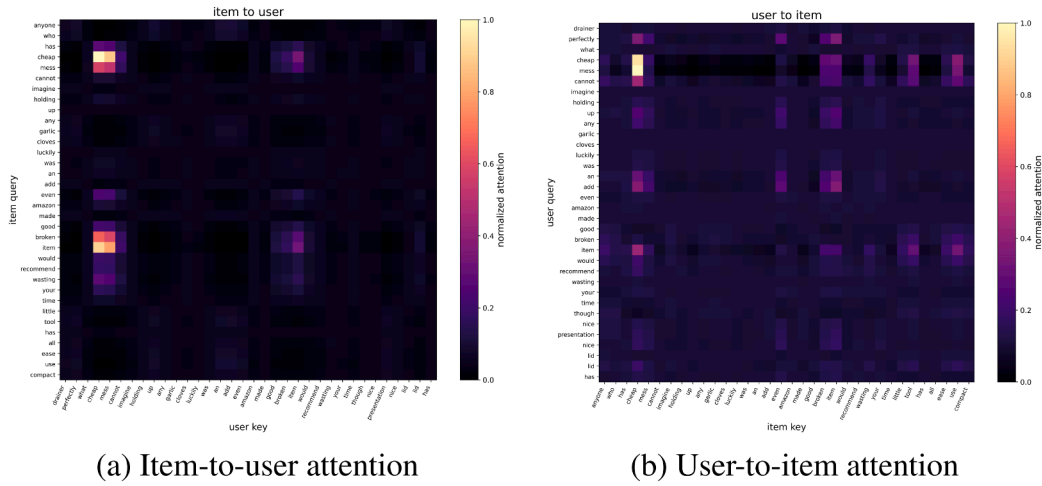


Fig. 9. Visualization of the dual cross-attention mechanism in PAAN. Attention values are averaged across heads and normalized separately for each direction.

ing speed among the compared methods. KEGC and RGCL also maintain moderate computational costs, indicating that these baseline models are relatively efficient in terms of training time. In contrast, PAAN requires a longer average epoch time than these lightweight baselines. This is mainly because the proposed position-aware excitation module introduces additional gating operations, and the dual cross-attention mechanism performs bidirectional token-level alignment between user and item review representations. These components improve the model's ability to capture position-sensitive semantic signals and fine-grained user-item interactions, but they also increase the training cost.

Nevertheless, PAAN achieves the best overall recommendation effectiveness rank, indicating that the additional computational cost brings consistent performance gains. Compared with APH, PAAN also maintains a substantially lower training time while achieving better prediction performance. These results suggest that PAAN provides a favorable trade-off between recommendation accuracy and computational efficiency. However, its efficiency is still lower than that of lightweight CNN-based or graph-based baselines, which remains a limitation of the current framework and motivates future optimization of the model structure.

5. Conclusions

In this paper, we propose a novel Position-aware Attentional Neural Network (PAAN) for review-based recommendation, which exploits structural and semantic information in user and item reviews through positional sequence excitation and dual cross-attention. Specifically, the position-aware excitation module recalibrates sequential representations by emphasizing sentiment-rich positions, while the dual cross-attention mechanism captures bidirectional dependencies between user and item review representations. The resulting review features are further fused with ID embeddings for rating prediction. Extensive experiments on multiple public datasets demonstrate the effectiveness and generalization ability of PAAN, showing that explicit position-aware modeling is useful for capturing informative sentiment patterns in sparse review data.

Although PAAN achieves strong recommendation performance, its training efficiency is still not optimal due to the additional position-aware excitation and dual cross-attention components. In addition, since PAAN adopts fixed-length review representations, very long reviews are truncated during preprocessing, which may lead to the loss of some fine-grained preference cues contained in lengthy review texts. Moreover, by explicitly modeling sentiment-sensitive review positions, PAAN may provide useful insights for future fairness-aware recommendation studies, where user preferences expressed in textual feedback can be consid-

ered more transparently beyond sparse rating records. In future work, we will also further optimize the computational efficiency of PAAN and explore graph-enhanced extensions, such as heterogeneous user-item-review graphs and dynamic interaction graphs, to capture higher-order relational dependencies beyond the current ID-based feature fusion.

CRedit authorship contribution statement

Yuanpeng Jiang: Writing – original draft, Visualization, Validation, Software, Formal analysis; **Teng Long:** Writing – review & editing, Supervision, Conceptualization; **Zhangbing Zhou:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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